

Q1 - 25 July - Shift 2

$$2\sin\left(\frac{\pi}{22}\right)\sin\left(\frac{3\pi}{22}\right)\sin\left(\frac{5\pi}{22}\right)\sin\left(\frac{7\pi}{22}\right)\sin\left(\frac{9\pi}{22}\right)$$

is equal to

(A) $\frac{3}{16}$

(B) $\frac{1}{16}$

(C) $\frac{1}{32}$

(D) $\frac{9}{32}$

Space for your notes:

Q2 - 27 July - Shift 2

Let $S = \left\{ \theta \in \left(0, \frac{\pi}{2}\right) : \sum_{m=1}^9 \sec\left(\theta + (m-1)\frac{\pi}{6}\right) \sec\left(\theta + \frac{m\pi}{6}\right) = -\frac{8}{\sqrt{3}} \right\}$

Then

(A) $S = \left\{ \frac{\pi}{12} \right\}$

(B) $S = \left\{ \frac{2\pi}{3} \right\}$

(C) $\sum_{\theta \in S} \theta = \frac{\pi}{2}$

(D) $\sum_{\theta \in S} \theta = \frac{3\pi}{4}$

Space for your notes:

Q3 - 28 July - Shift 2

Let $S = \left[-\pi, \frac{\pi}{2} \right) - \left\{ -\frac{\pi}{2}, -\frac{\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4} \right\}$. Then the

number of elements in the set

$$A = \left\{ \theta \in S : \tan \theta (1 + \sqrt{5} \tan(2\theta)) = \sqrt{5} - \tan(2\theta) \right\}$$

is _____

Space for your notes:

Answer Key

Q1 (B)

Q2 (C)

Q3 (5)

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Q1 (B)

(B)

Q2 (C)

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$$\text{Let } \alpha = \theta + (m-1)\frac{\pi}{6}$$

$$\& \beta = \theta + m\frac{\pi}{6}$$

$$\text{So, } \beta - \alpha = \frac{\pi}{6}$$

$$\text{Here, } \sum_{m=1}^9 \sec \alpha \cdot \sec \beta = \sum_{m=1}^9 \frac{1}{\cos \alpha \cdot \cos \beta}$$

$$= 2 \sum_{m=1}^9 \frac{\sin(\beta - \alpha)}{\cos \alpha \cdot \cos \beta} = 2 \sum_{m=1}^9 (\tan \beta - \tan \alpha)$$

$$= 2 \sum_{m=1}^9 \left(\tan \left(\theta + m\frac{\pi}{6} \right) - \tan \left(\theta + (m-1)\frac{\pi}{6} \right) \right)$$

$$= 2 \left(\tan \left(\theta + \frac{9\pi}{6} \right) - \tan \theta \right) = 2(-\cot \theta - \tan \theta) = -\frac{8}{\sqrt{3}}$$

(Given)

$$\therefore \tan \theta + \cot \theta = \frac{4}{\sqrt{3}}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{3}} \text{ or } \sqrt{3}$$

$$\text{So, } S = \left\{ \frac{\pi}{6}, \frac{\pi}{3} \right\}$$

$$\sum_{\theta \in S} \theta = \frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$$

Q3 (5)

$$\tan \theta + \sqrt{5} \tan 2\theta \tan \theta = \sqrt{5} - \tan 2\theta$$

$$\tan 3\theta = \sqrt{5}$$

$$\theta = \frac{n\pi}{3} + \frac{\alpha}{3}; \tan \alpha = \sqrt{5}$$

Five solution

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